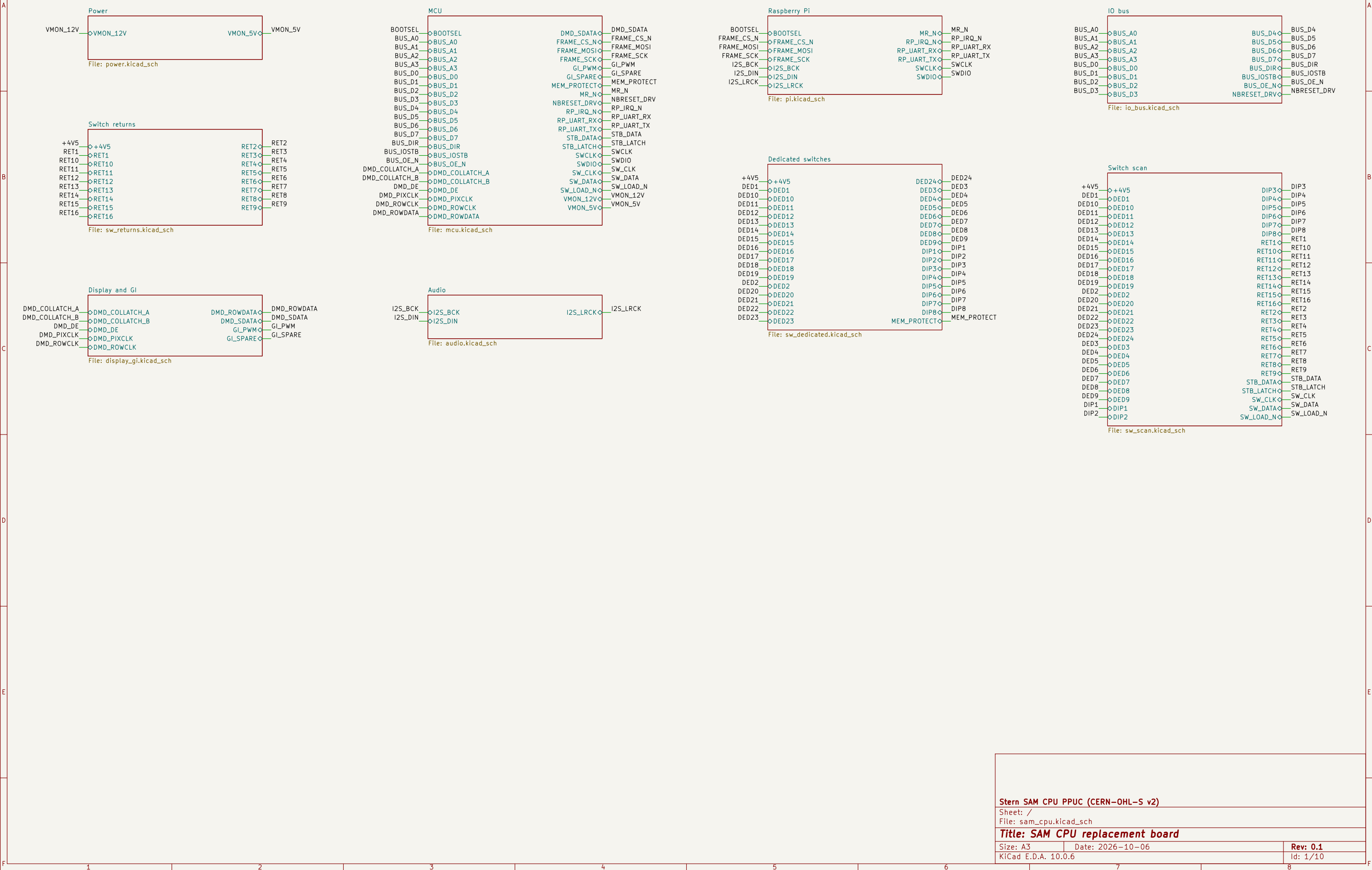


SAM CPU replacement board, draft 0.1, drawn from docs/ARCHITECTURE.md.
A Raspberry Pi 4 / CM4 runs PinMAME + ppuc. The RP2354B drives the original IO power driver board on J9 and scans the switches. Connector names and pinouts follow the original CPU/Sound board 520–5246–00.
Licence: CERN–OHL–S v2.



Stern SAM CPU PPUC (CERN–OHL–S v2)

Sheet: /
File: sam_cpu.kicad_sch

Title: SAM CPU replacement board

Size: A3 Date: 2026–10–06

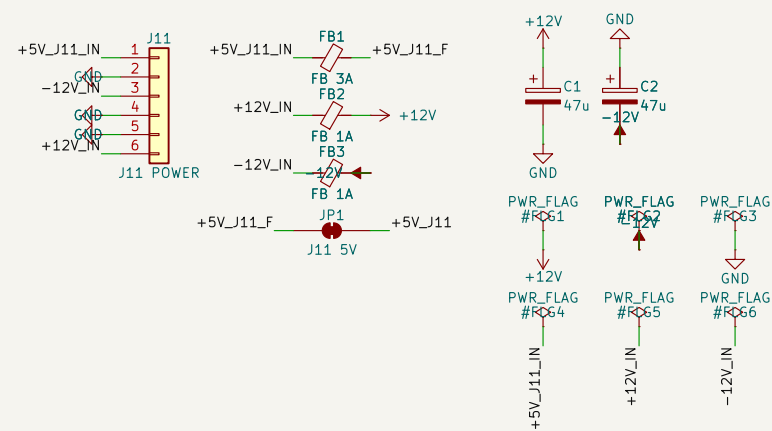
KiCad E.D.A. 10.0.6

Rev: 0.1

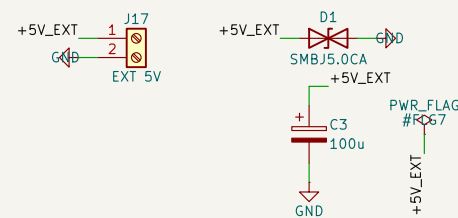
Id: 1/10

J11 keeps the original CPU board pinout (IO board J16 harness): +5 V, +−12 V and ground. External +5 V on J17 has priority: U2 turns the J11 path off whenever +5V_EXT is present (EXT_PRESENT high). LTC4412 + A03401A ideal diodes stand in for the TPS2121 of ARCHITECTURE.md 9.3 (library part, about 4 A per path). JP1 (bridged by default) disconnects the J11 +5 V completely; leaving J17 unplugged forces J11.

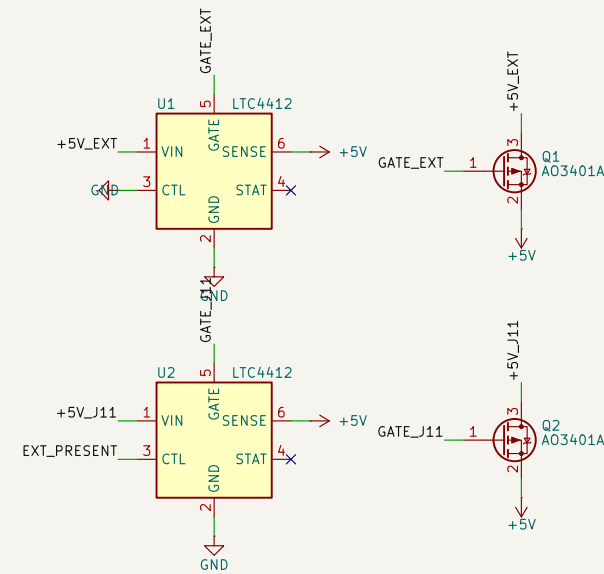
J11: power from the IO board (J16 harness)



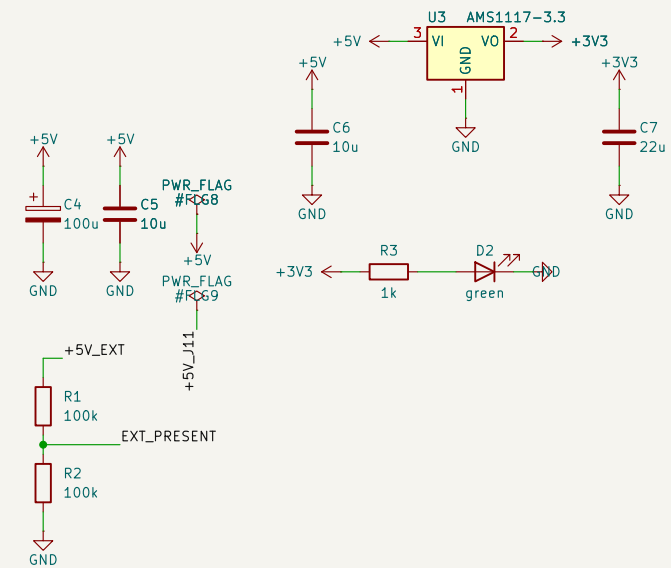
J17: external +5 V (priority)



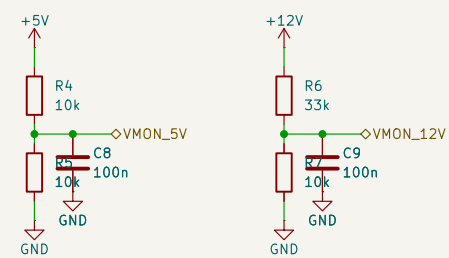
Ideal-diode power mux



3.3 V regulator

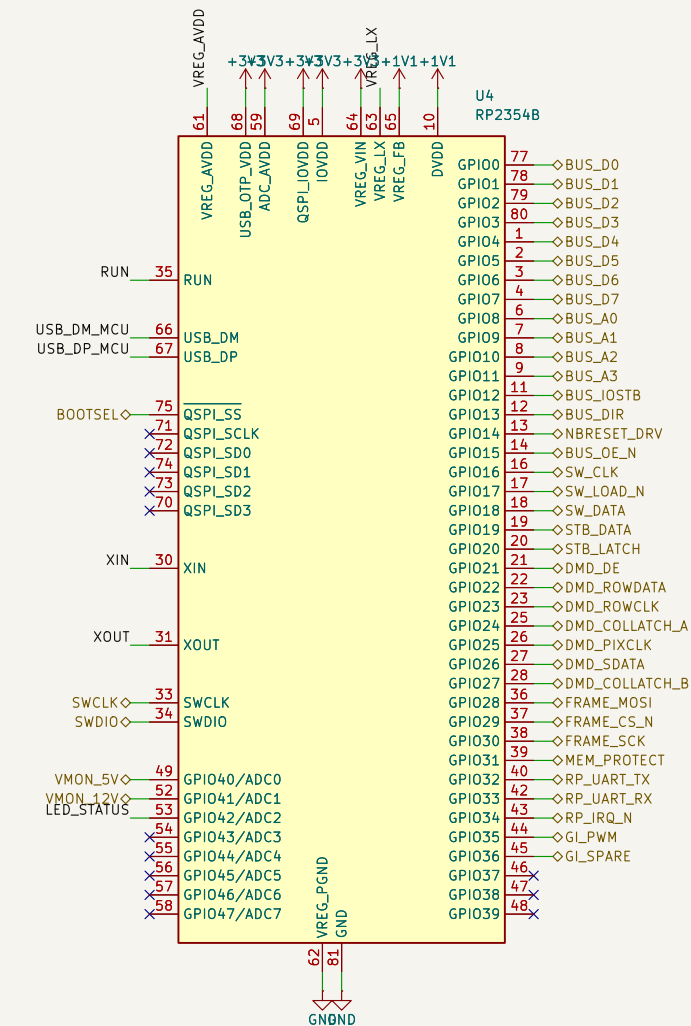


Supply monitors (RP2354B ADC)

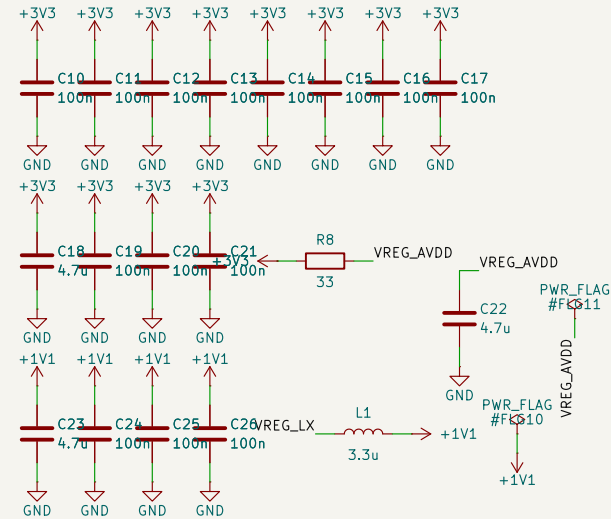


GPIO map follows ARCHITECTURE.md 9.2: PIO0 bus on GPIO0-15, PIO1 switch chain on GPIO16-20, PIO2 DMD on GPIO21-27. GPIO31 reads the coin door memory protect input (J2 pin 10), which the doc does not list yet.
Core supply from the internal switching regulator: VREG_LX -> L1 -> DVDD (1.1 V), VREG_AVDD through 33 R / 4.7 uF.
Check these values against the RP2350 hardware design guide before layout.

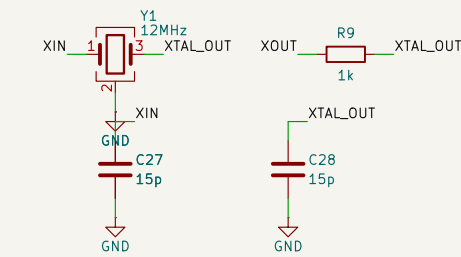
RP2354B (2 MB flash in the package)



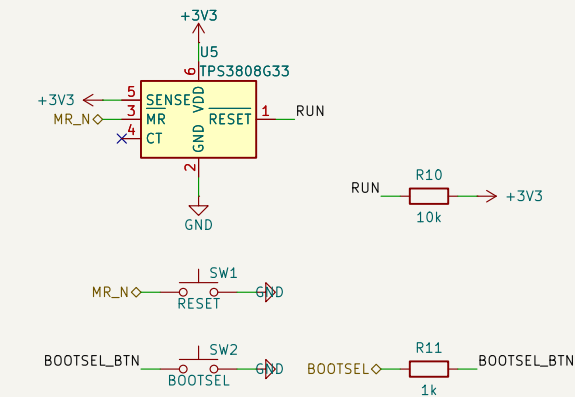
Decoupling and core regulator



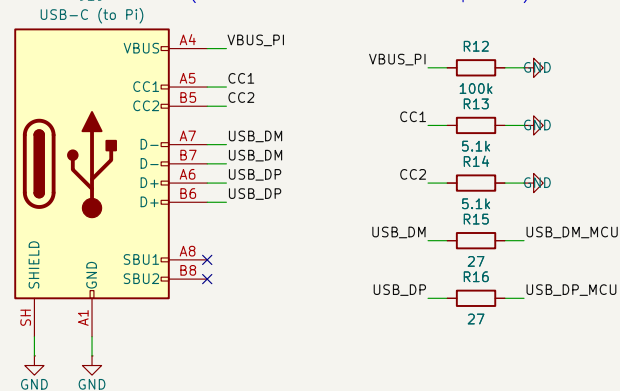
12 MHz crystal



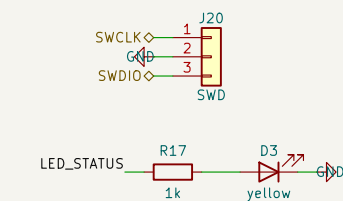
Reset supervisor, RESET and BOOTSEL



USB to the Pi (CDC link + firmware update)

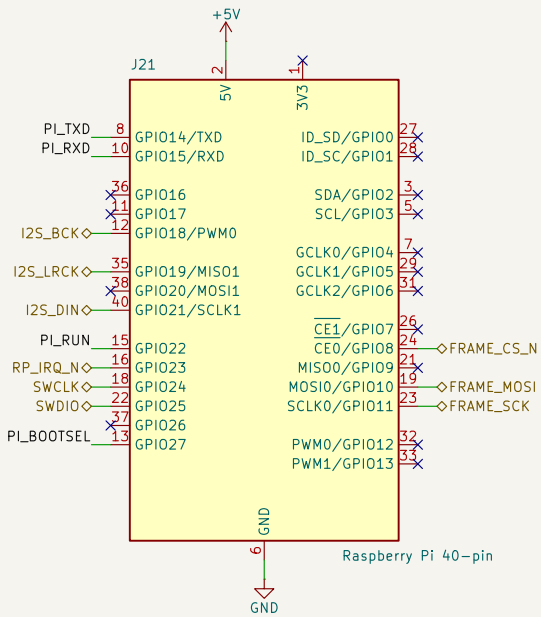


SWD header and status LED

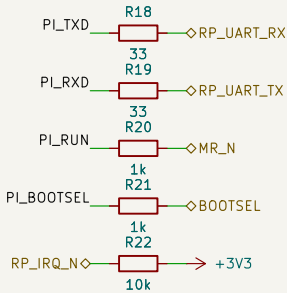


Only GPIOs a CM4 has on the same numbers (ARCHITECTURE.md 9.1). The Pi 3V3 pins stay unconnected.
The Pi is powered from the board +5 V through pins 2 and 4: never plug the Pi USB-C supply at the same time.
Link: UART0 (GPIO14/15), SPI0 for DMD frames, I2S to the DAC, RUN / BOOTSEL / SWD to the RP2354B, IRQ back.

Pi header



Pi to RP2354B link



Stern SAM CPU PPUC (CERN-OHL-S v2)

Sheet: /Raspberry Pi/
File: pi.kicad_sch

Title: Raspberry Pi 4 / CM4 40-pin header

Size: A3 Date: 2026-10-06

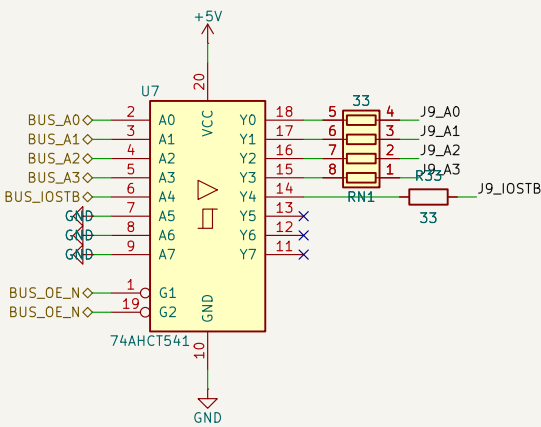
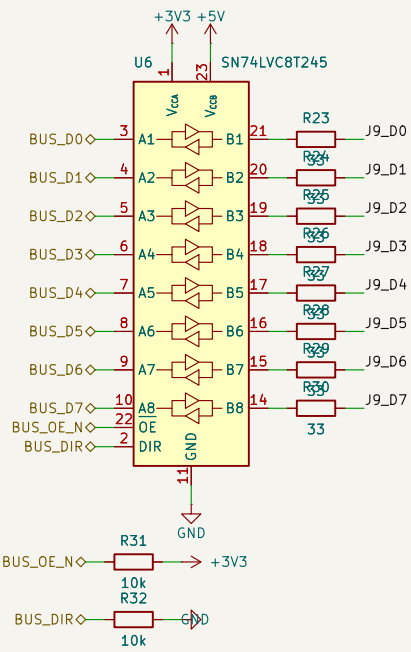
KiCad E.D.A. 10.0.6

Rev: 0.1

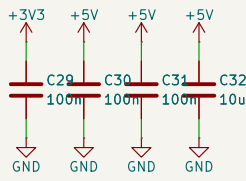
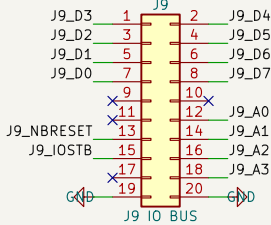
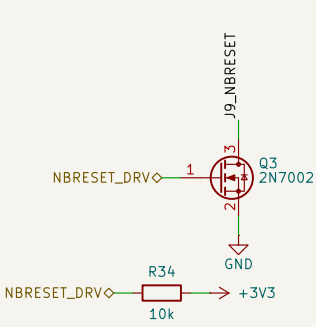
Id: 4/10

ARCHITECTURE.md 3.1–3.2. The doc calls this connector J1 (its name on the IO board); on the CPU board it is J9. Both buffers are disabled at power-up (BUS_OE_N pulled high) and NBRESET is held low until firmware releases it. Series resistors on the J9 side for the ribbon cable. Unused 74AHCT541 inputs tied low.

Data: SN74LVC8T245 (A = 3.3 V side, B = 5 V side) and IOSTB: 74AHCT541 at 5 V

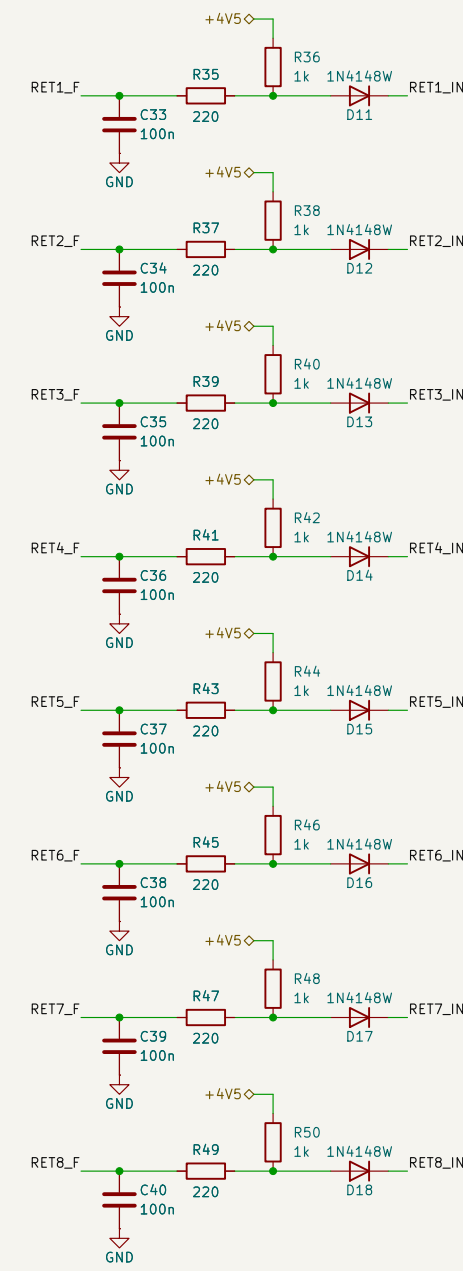


NBRESET: open drain, IO board to CPU board until released

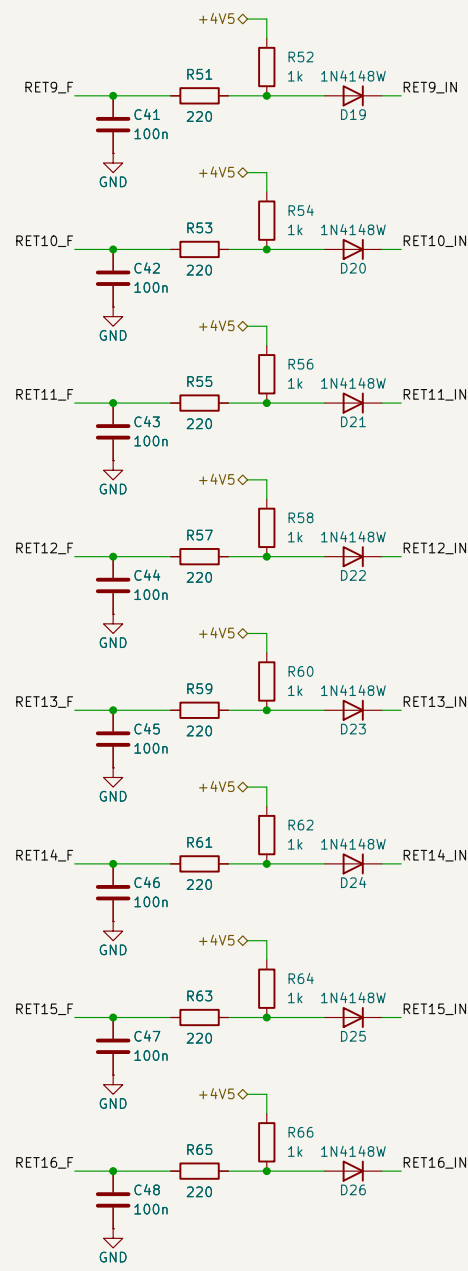


Per return (ARCHITECTURE.md 5.2–5.3): series diode from the connector, 1 k pull-up to +4.5 V, 220 R + 100 nF, LM339 against VREF (2.25 V). Output 1 = open, 0 = closed, as the ROM expects. Connector pin to return order follows the original drawing: verify with a meter before relying on it.

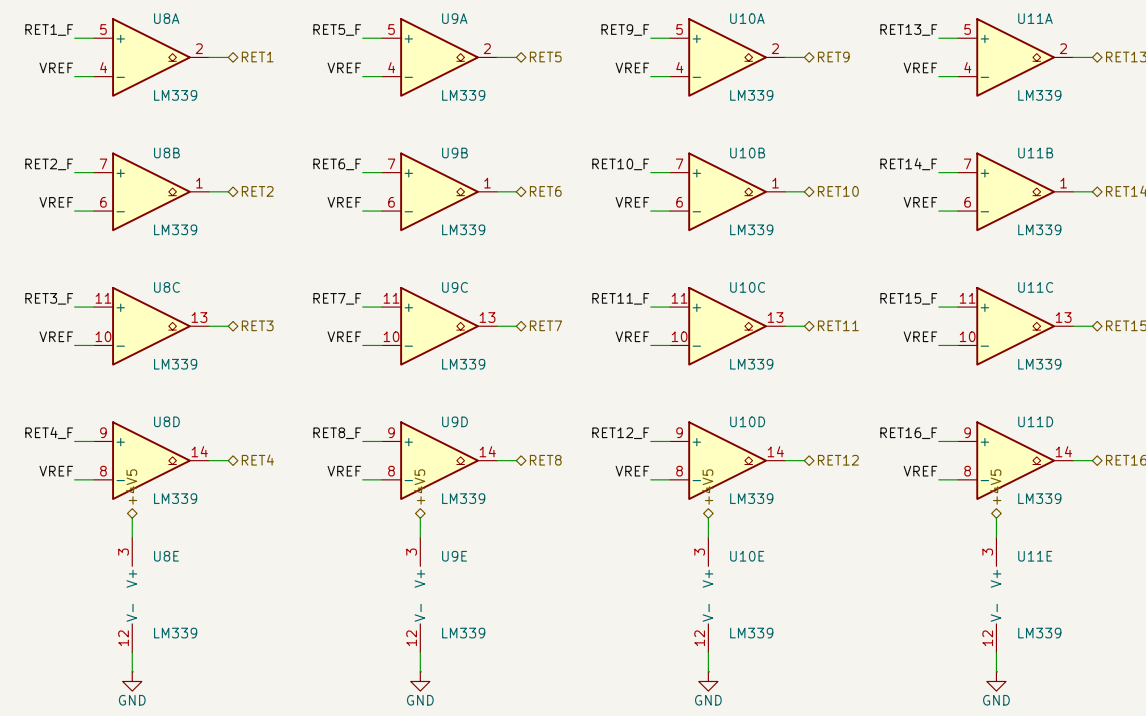
Returns 1–8 (J6)



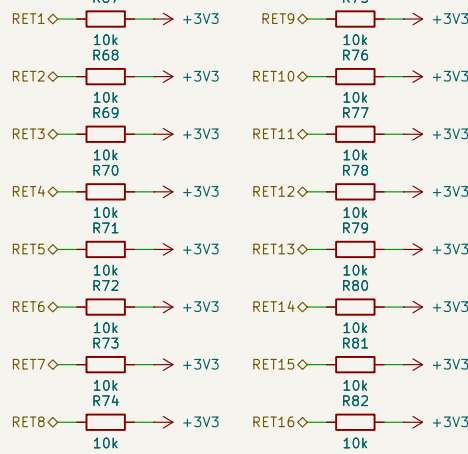
Returns 9–16 (J12)



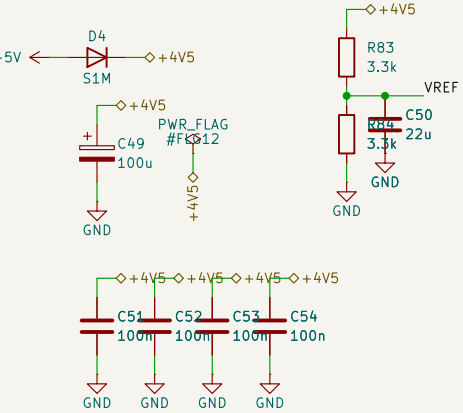
LM339 comparators (outputs to the shift chain)



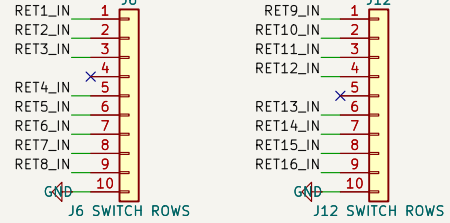
Comparator pull-ups (3.3 V logic)



+4.5 V switch supply and VREF

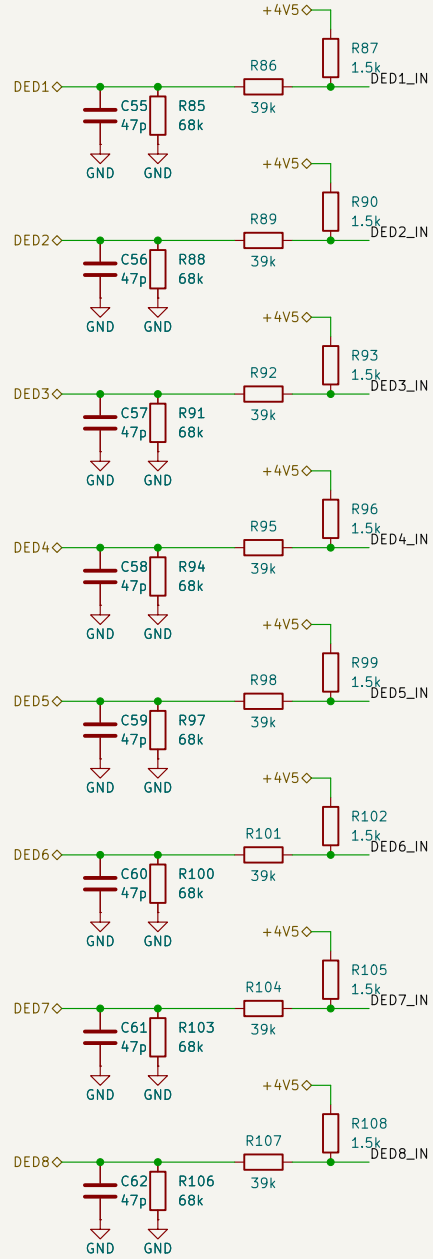


J6 and J12: switch rows (returns)

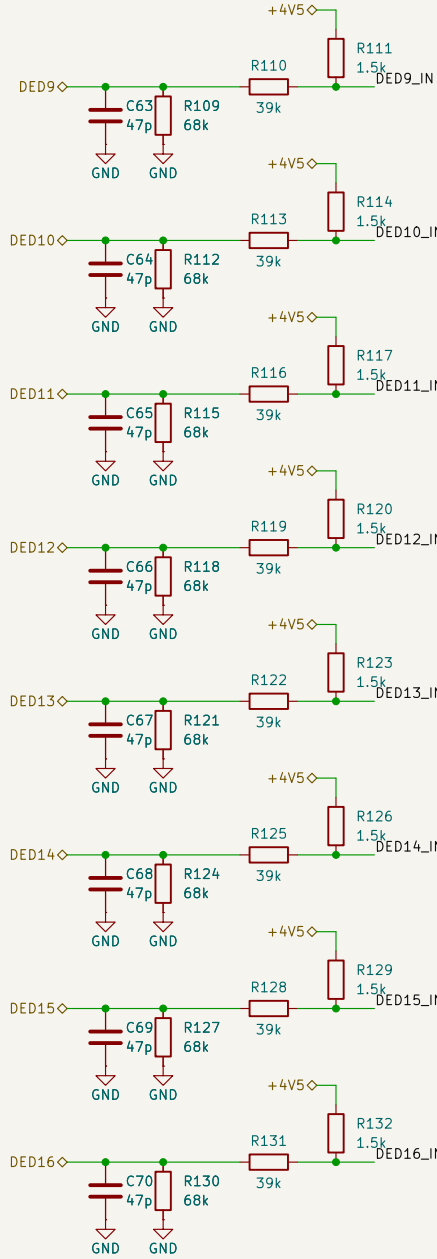


Per input: 1.5 k pull-up to +4.5 V (switch wetting current) and the original 39 k + 47 pF filter, plus 68 k to ground so the 3.3 V 74HC165 sees 2.8 V (open) or 0 V (closed). Connector pin to D-number order follows the original drawing: verify it. Memory protect (J2 pin 10) uses the same filter with the original 4.7 k pull-up and goes to RP2354B GPIO31.

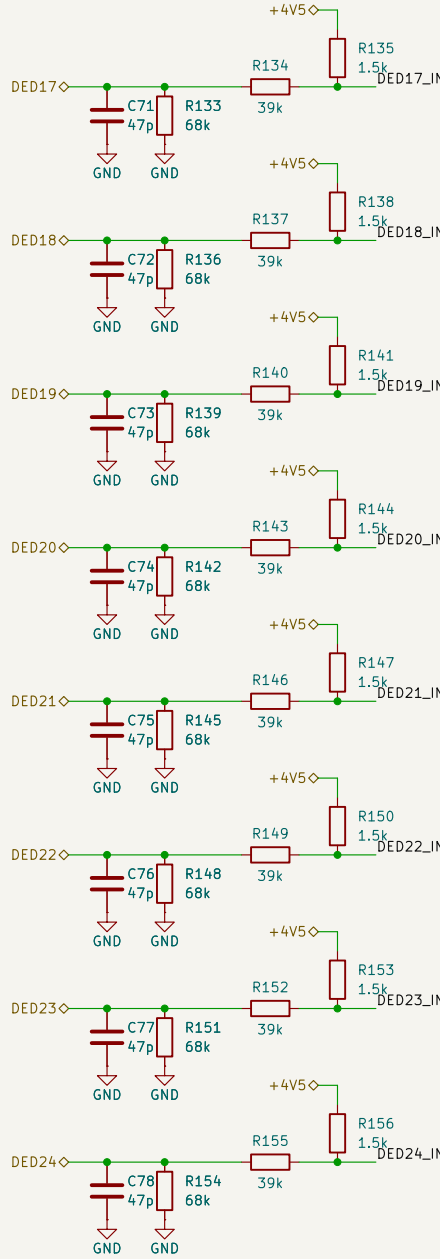
D1–D8 (J2)



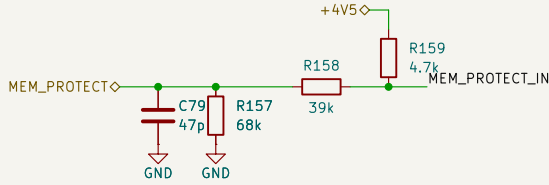
D9–D16 (J3)



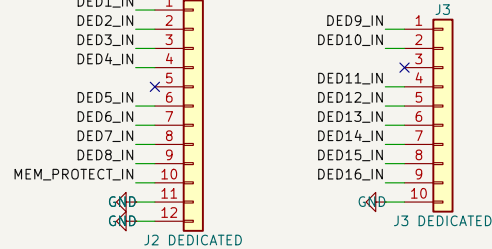
D17–D24 (J13)



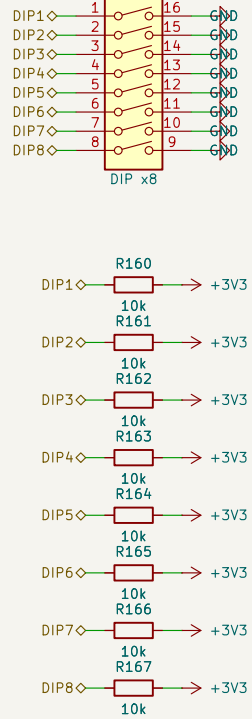
Memory protect (J2 pin 10)



Connectors J2, J3, J13

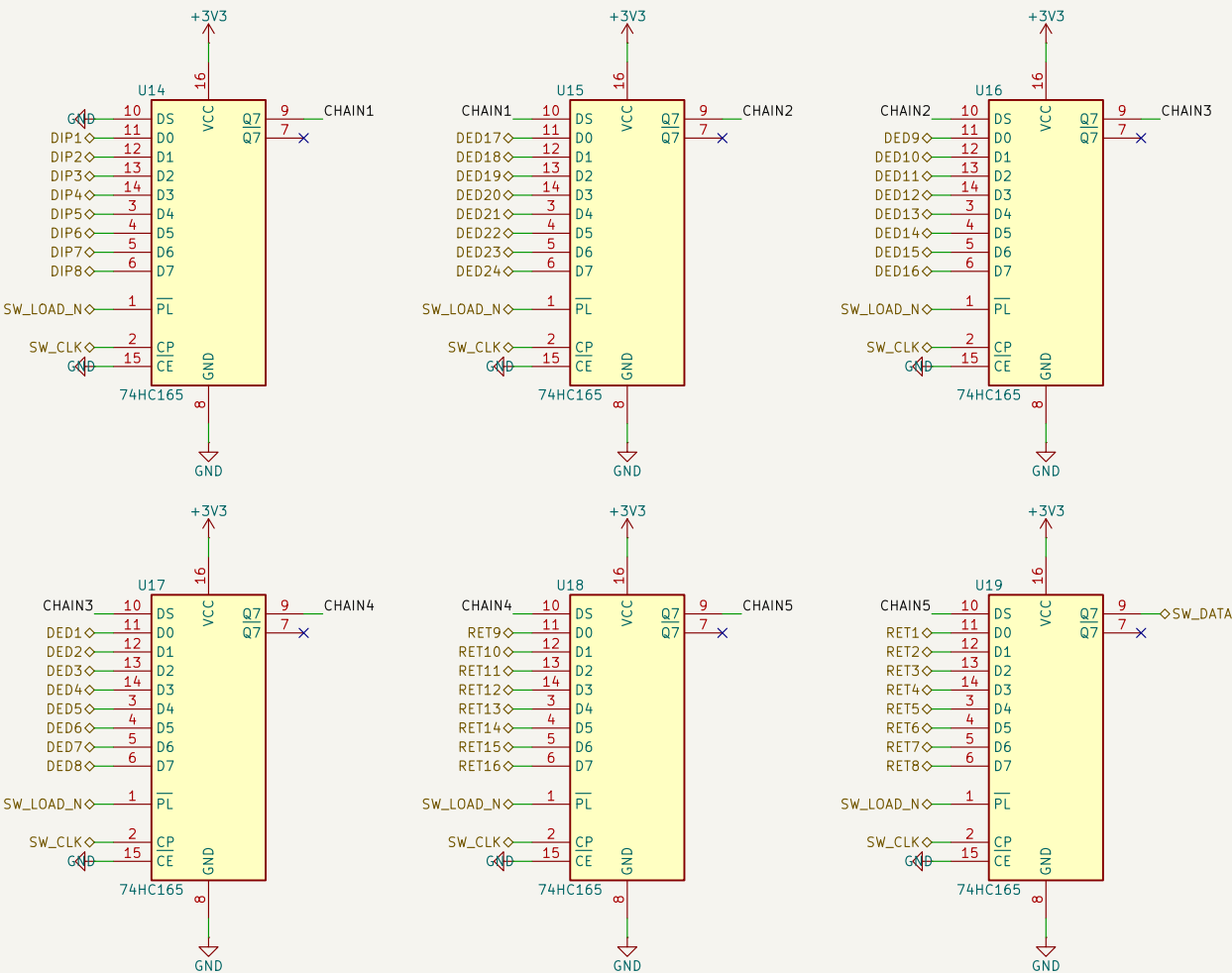


DIP switches (D25–D32)

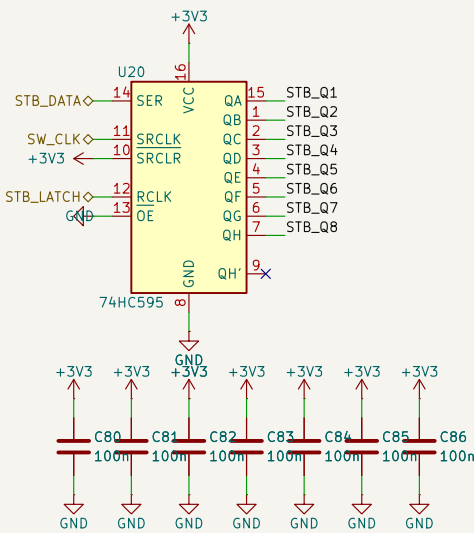


One shift clock for both registers: PIO1 shifts 48 inputs in while the next strobe pattern shifts into the 595 (ARCHITECTURE.md 5.3–5.4).
First bits out of SW_DATA: U19 RET1–8, U18 RET9–16, U17 D1–8, U16 D9–16, U15 D17–24, U14 DIP1–8.
Strobes: 595 high -> MMBT3904 on -> strobe pulled low (active), with the original 1 k pull-up to +4.5 V and 100 nF.

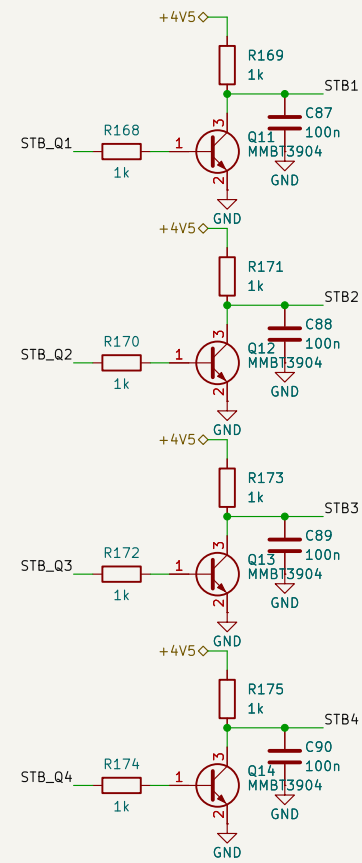
74HC165 input chain (3.3 V)



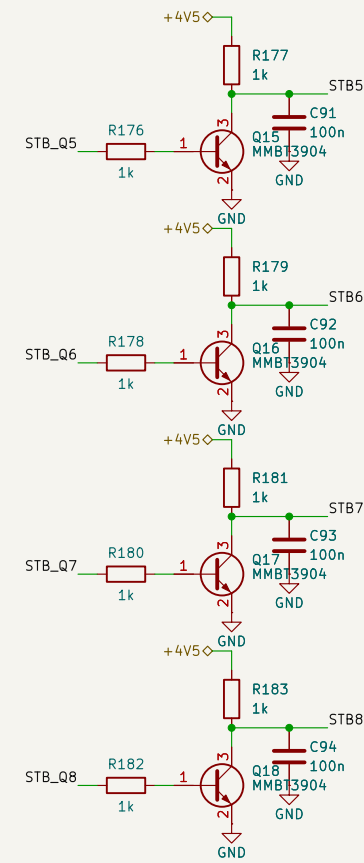
74HC595 strobe register



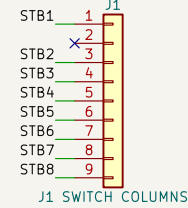
Strobe drivers 1–4



Strobe drivers 5–8

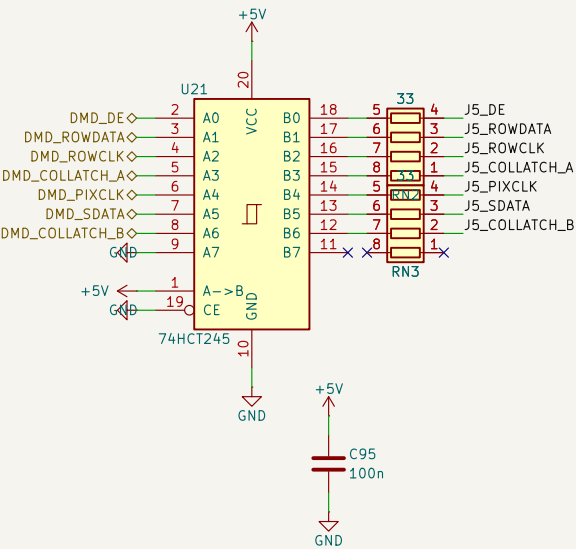


J1: switch columns (strobes)

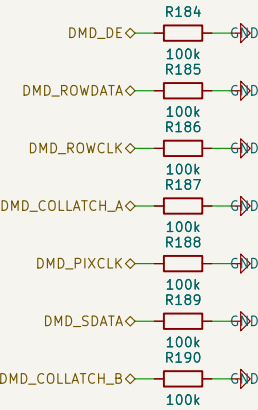


J5 pinout read from the original drawing (ARCHITECTURE.md 8): signals on odd pins, ground on even pins. Confirm with a meter (Q18).
The 100 k pull-downs keep the DMD lines low while the RP2354B boots.
J18 is the default GI dimmer link of ARCHITECTURE.md 10.2: +5 V, GND, GL_PWM, GL_SPARE.

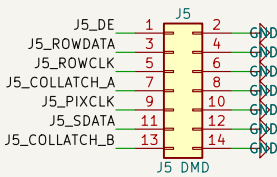
74HCT245 at 5 V (A to B only)



Pull-downs



J5: DMD (2x7)



J18: GI dimmer board link



Stern SAM CPU PPUC (CERN-OHL-S v2)

Sheet: /Display and GI/
File: display_gi.kicad_sch

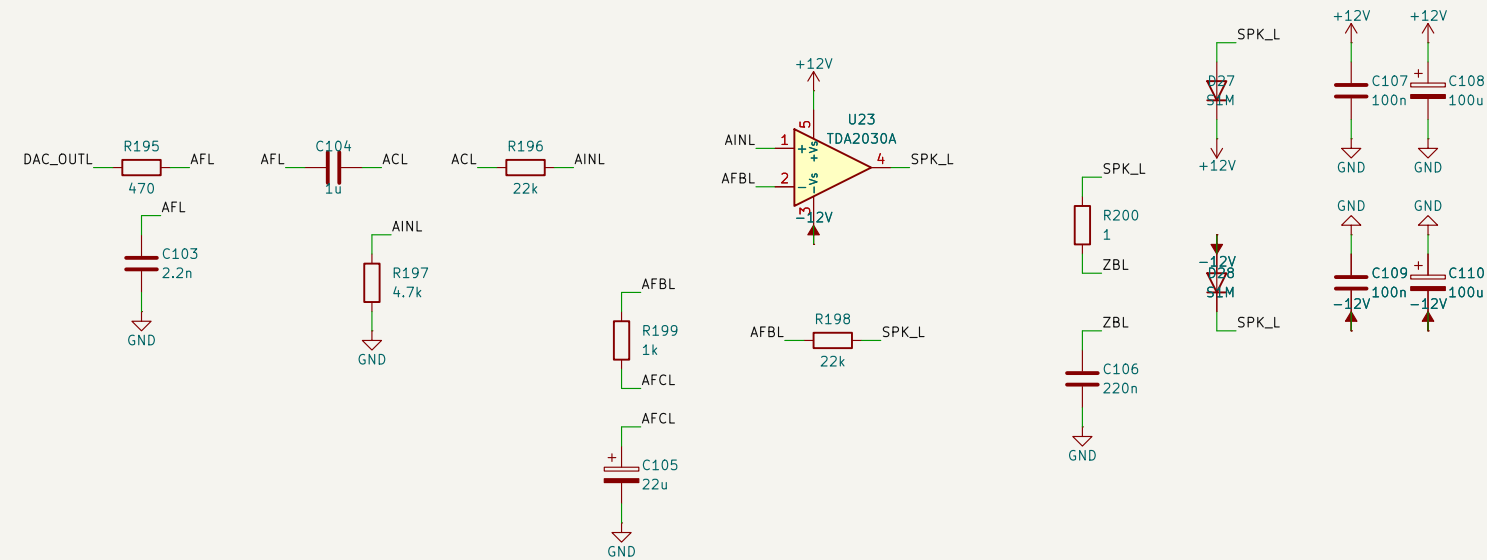
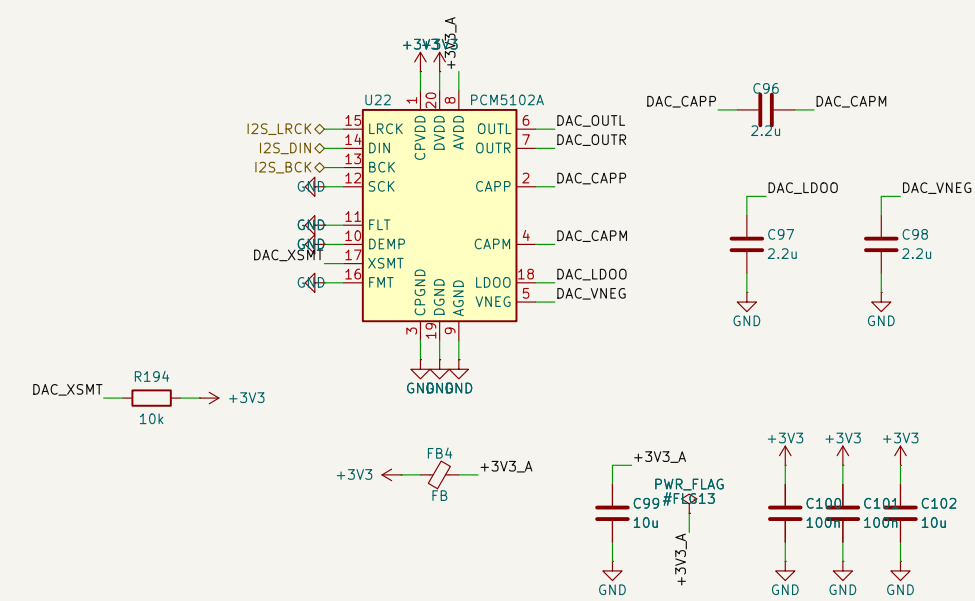
Title: J5 original DMD driver and GI dimmer header

Size: A3	Date: 2026-10-06	Rev: 0.1
KiCad E.D.A. 10.0.6		Id: 9/10

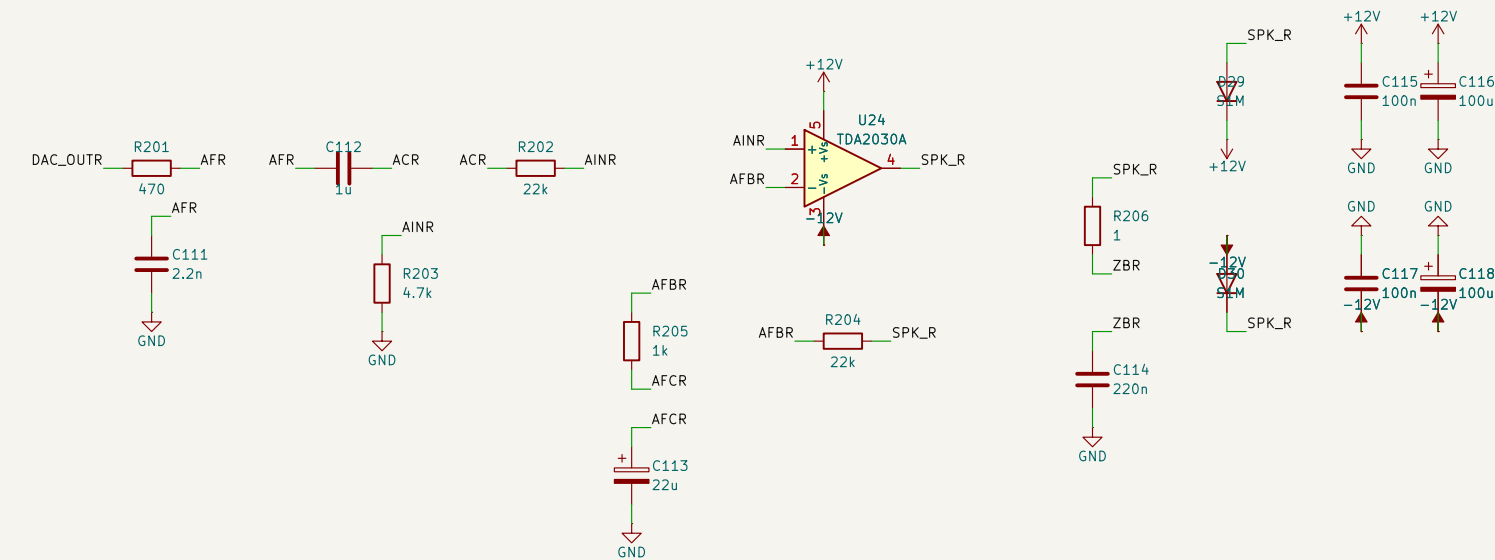
Same structure as the original (DAC + two TDA2030A on +12 V, speakers on J10). Volume stays digital (ROM / PinMAME). Gain: $22k / 4.7k$ input divider $\times (1 + 22k / 1k) =$ about 4 overall. Starting values, to check on the bench.

PCM5102A DAC (PLL from BCK, I2S format)

Left amplifier



Right amplifier



J10: speakers (original pinout)

